

Mahanth Sai S

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SUMMARY

- Data Engineer with 4+ years of experience designing and maintaining robust data pipelines across AWS and Azure platforms, focusing on data quality, security, and database management, while ensuring data integrity and attention to detail
- Expert in deploying scalable ETL/ELT solutions using AWS Glue, Azure Data Factory, Apache Airflow, and Databricks, with an emphasis on data extraction and integration from ERP systems and data warehouses, enhancing business reporting
- Experienced in supporting both batch and real-time data processing workflows, ensuring data integrity and attention to detail, while maintaining data security across platforms
- Began Career by Architecting AWS-based data platforms for telecom and finance sectors, specializing in database design, management, and big data technologies, with a focus on infrastructure and security
- Proficient in Python, SQL, and PySpark for data cleaning, transformation, and automation, with knowledge of Java and information technology for enhanced data engineering, including data access controls
- Hands-on experience with Snowflake, Amazon Redshift, and Azure SQL for data warehousing and management of relational databases, with a focus on teamwork and collaboration
- Built insights-driven dashboards using Power BI, Amazon Quicksight, and Tableau, contributing to effective business reporting and data visualization, leveraging Microsoft Power Platform
- Delivered HIPAA-compliant, cost-optimized cloud solutions tailored to business needs, with a focus on data security, access controls, and the Microsoft Power Platform, while being aware of recruitment processes

SKILLS

Category	Technologies & Tools
Programming Languages	Python, SQL, T-SQL, TypeScript, Java, Scala, Shell Scripting, Bash
Data Processing Frameworks	PySpark, Apache Spark, Hadoop, MapReduce, Pandas, NumPy
Cloud Platforms - AWS	Glue, Kinesis, Lambda, Redshift, EMR, S3, CloudWatch, CodePipeline
Cloud Platforms - Azure	Data Factory, Data Lake, Event Hubs, Databricks, SQL Database, DevOps
ETL/ELT Tools	AWS Glue, Azure Data Factory, SSIS, Apache Airflow
Data Warehouses	Snowflake, Amazon Redshift, Azure SQL Database
Databases	SQL Server, PostgreSQL, MongoDB, Cassandra
Big Data Tools	Apache Hive, Apache Kafka, Hadoop Ecosystem
Data Modeling	Star Schema, Snowflake Schema, Dimensional Data Modeling
Visualization Tools	Tableau, Power BI, AWS QuickSight
DevOps & CI/CD	Docker, Kubernetes, Jenkins, Azure DevOps, AWS CodePipeline
Data Quality & Governance	Data Profiling, Data Validation, AWS Glue Data Catalog
Security & Compliance	IAM, KMS, RBAC, S3 Encryption, HIPAA, FHIR, HL7
Monitoring & Alerting	CloudWatch, Pipeline Monitoring, Performance Optimization
Integration Methods	REST APIs, SFTP/FTPS, API Gateway, Logic Apps
Additional Platforms	Microsoft Power Platform, SharePoint, Django

PROFESSIONAL EXPERIENCE

Data Engineer, CVS Health, Providence, United States

Aug 2023 — Present

- Developed ETL pipelines in Azure Data Factory, Logic Apps, and SSIS for ingesting healthcare data from SQL Server, Blob Storage, and Hive into Azure SQL, ensuring high data quality and integrity while managing data warehouse infrastructure and focusing on maintenance.
- Automated ingestion of REST APIs using ADF and Python, with robust error handling via T-SQL. Built advanced Power BI and Tableau dashboards for real-time monitoring of healthcare KPIs, enhancing operational decision-making, data visualization, and attention to detail while supporting business reporting needs.

- Ensured compliance with HIPAA, FHIR, HL7, and CDA data standards, focusing on robust data security measures and database management, while implementing data access controls.
- Designed star/snowflake schemas using Power Designer to support OLAP reporting and optimize database design, improving data warehousing solutions and ensuring data modeling excellence.
- Implemented CI/CD pipelines in Azure DevOps, integrating automated Python unit and integration tests to ensure code reliability.
- Collaborated with cross-functional teams to enhance data governance and optimize healthcare data workflows, aligning with business reporting requirements and ensuring effective teamwork and communication.
- Orchestrated incremental and full data loads using watermarking in ADF, improving data latency and pipeline performance across multiple healthcare systems, ensuring efficient data extraction and processing with attention to detail.
- Performed data profiling and quality assurance using Databricks and T-SQL to ensure consistency and accuracy of PHI in clinical datasets, supporting data engineering excellence and data integrity with robust problem-solving.
- Integrated and secured data exchange via SFTP/FTPS and API connectors, improving interoperability across healthcare systems and ensuring data access security, focusing on information technology standards and compliance.
- Implemented RBAC and encryption protocols in Azure to safeguard patient data, reinforcing enterprise-wide HIPAA compliance and data security with attention to detail and effective data governance.

Data Engineer, Sutherland Global Services, Hyderabad, India

Sep 2021 — Dec 2022

- Created robust batch and real-time ingestion workflows using AWS Glue, EMR, and Apache Airflow, ensuring efficient data extraction and processing with a focus on data quality. Processed 10TB+ datasets in Amazon S3 using PySpark, cutting load times by 35%.
- Modeled financial data marts in Redshift, powering dashboards for 200+ finance analysts and enhancing data visualization capabilities with effective data reporting.
- Built and deployed cost-effective analytics dashboards in AWS Quicksight, supporting business reporting and decision-making with advanced data visualization techniques.
- Automated CI/CD workflows via AWS CodePipeline, enhancing deployment speed. Integrated real-time feeds using Lambda and Kinesis, enabling near real-time decisioning.
- Led collaboration with product managers to gather KPIs and develop scalable reporting pipelines, ensuring data quality and integrity with a focus on teamwork.
- Secured data ingestion and exchange via SFTP/API Gateway, applying IAM, KMS, and S3 encryption policies to ensure compliance, data privacy, and enterprise-grade data security with attention to detail.
- Optimized Redshift performance using Spectrum, materialized views, and WLM tuning, resulting in 25% faster query response times and reduced storage costs.

Data Engineer, BSNL, Hyderabad, India

Oct 2020 — Aug 2021

- Designed and deployed ETL pipelines using AWS Glue, transforming multi-format telecom data into Redshift for national reporting and data warehousing with a focus on data security.
- Created visualization dashboards using AWS Quicksight to monitor customer churn and usage.
- Enabled ad-hoc insights using Athena, improving reporting agility for internal teams and supporting business reporting needs with a focus on data engineering.
- Maintained AWS Glue Data Catalog for improved data governance and discovery, ensuring data quality and integrity with a focus on data management.
- Migrated legacy SQL-based workloads to a serverless AWS architecture, enhancing data engineering capabilities with a focus on cloud infrastructure.
- Built monitoring and alerting using CloudWatch, improving pipeline uptime to 99.9%.
- Conducted deep root cause analysis of telecom data issues using Python scripts, supporting data engineering and problem-solving efforts with attention to detail.
- Refactored legacy ETL into modular Glue Workflows, optimizing cost and performance and ensuring efficient data extraction with a focus on maintenance.
- Secured data workflows using IAM roles and S3 bucket policies for access control, ensuring robust data security and compliance with attention to detail.

EDUCATION

University of Dayton, Dayton, United States - Master's, Computer Science

Jan 2023 — Aug 2024